SET 18: PRODUCT INVENTORY MANAGEMENT

Dataset	Product Inventory
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

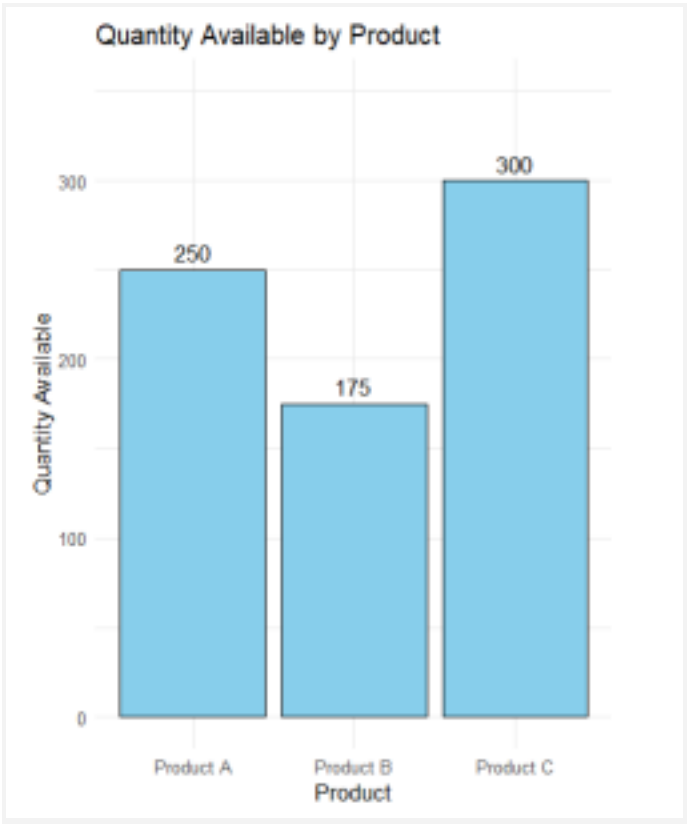
PROGRAM 1: Bar Chart of Product Quantity

Question: Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset
product <- c("Product A", "Product B", "Product C")
quantity <- c(250, 175, 300)

# Bar Chart
bp <- barplot(quantity,
 names.arg = product,
 main = "Quantity Available by Product",
 xlab = "Product",
 ylab = "Quantity Available",
 col = "skyblue",
 ylim = c(0, 350))

# Label each bar
text(bp, quantity, labels = quantity, pos = 3)
```



Output 1: Product Quantity Bar Chart

Observation: The bar chart shows the available quantity of each product. Product C has the highest stock (300), followed by Product A (250), and Product B has the lowest stock (175).

PROGRAM 2: Stacked Bar Chart by Product Category

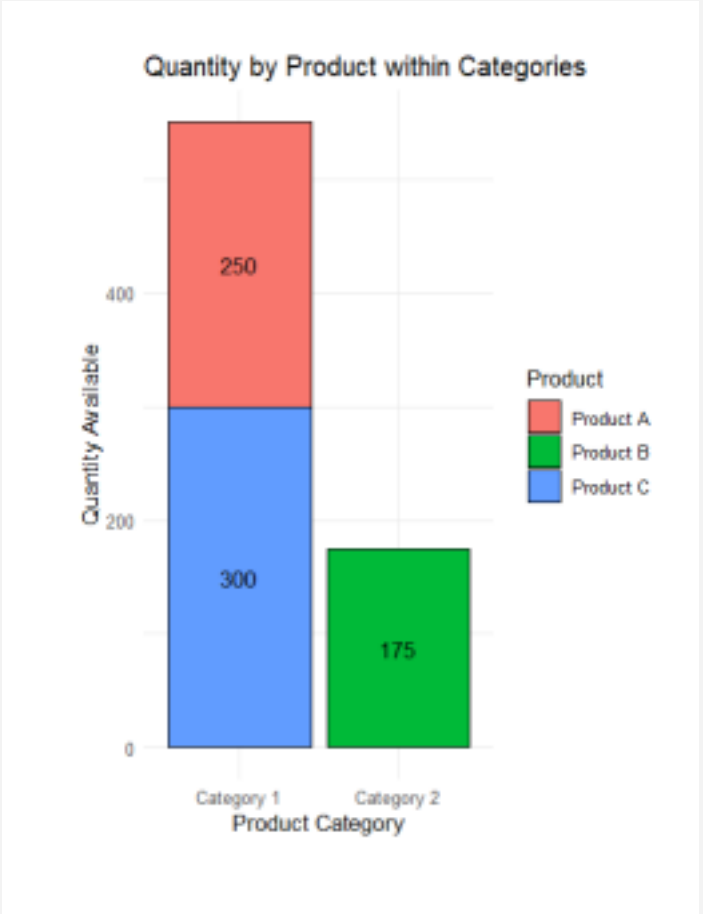
Question: Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Dataset (the Category column is assumed, as the dataset does not give
one)
inventory <- data.frame(
 Product = c("Product A", "Product B", "Product C"),
 Category = c("Category 1", "Category 2", "Category 1"),
Quantity = c(250, 175, 300)
)

# Quantity of each product within each category
qty_matrix <- xtabs(Quantity ~ Product + Category, data = inventory)

# Stacked Bar Chart
barplot(qty_matrix,
 main = "Quantity by Product within Categories",
 xlab = "Product Category",
 ylab = "Quantity Available",
 col = c("skyblue", "orange", "lightgreen"),
 ylim = c(0, 650),
```

```
 legend.text = rownames(qty_matrix),
 args.legend = list(x = "topright", bty = "n"))
```



Output 2: Stacked Bar Chart of Quantity by Category

Observation: Each bar represents a product category and is stacked by product. Category 1 holds 550 units (Product A 250 and Product C 300), while Category 2 holds 175 units (Product B).

PROGRAM 3: Product Inventory Data Table

Question: Build a table to show the inventory data for each product. Label the table elements.

```
# Product Inventory Dataset
inventory_data <- data.frame(
 Product_ID = c(1, 2, 3),
 Product_Name = c("Product A", "Product B", "Product C"),
Quantity_Available = c(250, 175, 300),
 Price = c(20, 15, 18)
)

# Display the table
print(inventory_data)
```

Product Inventory Data Table

Product_ID	Product_Name	Quantity_Available	Price	Category
1	Product A	250	20	Category 1
2	Product B	175	15	Category 2
3	Product C	300	18	Category 1

Output 3: Product Inventory Data Table

Observation: The table displays the Product ID, Product Name, Quantity Available and Price (in $) for each of the three products.

PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Tableau Steps

1. Open Tableau and connect to the Product Inventory dataset.
2. Verify the fields Product ID, Product Name, Quantity Available and Price (in $); add a Category field (Category 1 / Category 2) to the data source for the stacked bar chart. 3. Create a worksheet for the bar chart: drag Product Name to Columns and Quantity Available to Rows, then select Bar in the Marks card.
4. Add the title “Quantity Available by Product” and label the axes.
5. Create a second worksheet: drag Category to Columns, SUM(Quantity Available) to Rows and Product Name to Color to get the stacked bar chart.
6. Add the title “Quantity by Product within Categories” and label the axes. 7. Create a third worksheet and drag Product ID, Product Name, Quantity Available and Price (in $) into the view to form the inventory table.
8. Create a new Dashboard and add the bar chart, stacked bar chart and table. 9. Add the title “Product Inventory Dashboard”.
10. Use Category or Product Name as a filter (click “Use as Filter”) to make the dashboard interactive.
11. Test the dashboard by selecting a product/category and checking the linked views.



Output 4: Product Inventory Dashboard

Observation: The Tableau dashboard combines the quantity bar chart, stacked bar chart and inventory table, allowing interactive exploration of the inventory data.

Procedure

1. Create the product inventory data in R.
2. Prepare the quantity values and create the bar chart using barplot().
3. Prepare the category data, build a quantity matrix using xtabs(), and create the stacked bar chart using barplot().
4. Create a data frame containing Product ID, Product Name, Quantity Available and Price and display it as a table.
5. Open Tableau and import the Product Inventory dataset.
6. Create the quantity bar chart and the stacked bar chart in Tableau.
7. Create the inventory data table in Tableau.
8. Combine the three visualizations into an interactive dashboard and add a suitable title. 9. Test the dashboard using the available filter/interactivity.

Result

The Product Inventory data was successfully visualized using a bar chart, stacked bar chart, inventory data table, and an interactive Tableau dashboard.